

DATA ANALYSIS WITH R AND RUST

Descriptive Analytics and Data Analysis

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Outline

1 Descriptive Analytics

2 Data Analysis



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R for Descriptive Statistics

R Language

R is a **statistical programming language** designed for **data analysis**, **visualization**, and **statistical computing**, widely used in *academia*, *biostatistics*, and *data science*.

Core capabilities:

- Vectors and data frames as native first-class types.
- Built-in functions: `mean()`, `sd()`, `summary()`, `cor()`, and `lm()`.

Key packages:

- `ggplot2`: data visualization
- `shiny`: interactive web applications
- `tidyr`: data manipulation and cleaning
- `caret`: machine learning workflow management



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Key packages:

- **dplyr**: data manipulation and filtering grammar.
- **tidyr**: tidy data preparation.
- **readr**: reading and writing flat files.
- **writer**: writing flat files.
- **readxl**: reading Excel spreadsheets.
- **writexl**: writing Excel spreadsheets.
- **readr**: reading CSV files.
- **writer**: writing CSV files.
- **readr**: reading JSON files.
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- **readr**: reading Parquet files.
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Data Composition I

Data Types in a Dataset

Understanding **data composition** means identifying **what variable types** exist and **how they are distributed**, which determines which *statistical methods* are applicable.



Data Composition II

- **Numerical (Quantitative):**

- **Continuous:** any real value in a range (velocity, distance, temperature).
- **Discrete:** countable integers (asteroid count, events per day).

- **Categorical (Qualitative):**

- **Nominal:** no order (country, orbiting body name).
- **Ordinal:** ordered categories (low / medium / high hazard level).

- **Temporal:** time-series data with date/time indices; requires specific handling for trend and seasonality.

- **Boolean:** binary flags, e.g., `is_potentially_hazardous`.

- **Missing:** NaN / null values that must be identified and handled *before* any statistical analysis.



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Mean and Median

Mean (Arithmetic Average)

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Sensitive to **outliers**; best for **symmetric** distributions.

Median

The **middle value** when data is sorted in ascending order.

Robust to outliers; preferred for **skewed** distributions.



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Mode and Range

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The **most frequent** value in the dataset.

Useful for **categorical** and **discrete** data; a dataset can be **unimodal**, **bimodal**, or **multimodal**.

Range

$$R = x_{\max} - x_{\min}$$

Simplest measure of spread; **highly sensitive** to **extreme values**.



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Normal Data, Standard Deviation, and Variance

Normal (*Gaussian*) Distribution

A **symmetric**, **bell-shaped** continuous probability distribution fully described by its **mean** μ and **standard deviation** σ :

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right)$$

Variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Average squared deviation from the mean; in **squared units**.

Standard Deviation

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Same units as data; measures dispersion around $\bar{x}$.

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Skewness and Kurtosis

Skewness

$$\gamma_1 = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$

- $\gamma_1 = 0$: **symmetric** (normal).
- $\gamma_1 > 0$: **right-skewed**, long right tail.
- $\gamma_1 < 0$: **left-skewed**, long left tail.

Excess Kurtosis

$$\gamma_2 = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^4 - 3$$

- $\gamma_2 = 0$: **mesokurtic** (normal tails).
- $\gamma_2 > 0$: **leptokurtic**, heavy tails.
- $\gamma_2 < 0$: **platykurtic**, light tails.

High **skewness** signals **outliers** or **non-normality**; high **kurtosis** indicates **extreme events** more frequent than in a normal distribution — critical for **financial** and **risk analysis**.

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Histograms and Data Distribution I

Histogram

A **histogram** divides data into contiguous **bins** and plots the **frequency** or **density** per bin, revealing the *shape*, *spread*, and *modality* of the data distribution.



Histograms and Data Distribution [II]

- **Bin width:** too narrow $\Rightarrow$ **noisy**; too wide $\Rightarrow$ **over-smoothed**.
Sturges' rule: $k = \lceil 1 + \log_2 n \rceil$ bins for n observations.
- **Unimodal:** **one peak** — possibly normal. **Bimodal:** **two peaks** — suggests a **mixture** of populations.
- **Density plot (KDE):** smooth **continuous estimate** of distribution using a **kernel function**; area under curve = 1.
- **Frequency vs. Density:**
 - **Frequency:** raw count of observations per bin.
 - **Density:** count $\div$ (total $\times$ bin width) — sums to 1.
- **Interpretation:** shape reveals **skewness**, **outliers**, and whether **normality** is a reasonable assumption.



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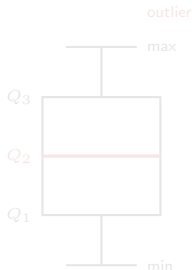


Box Plots and Data Confidence I

Box Plot (Box-and-Whisker)

A **boxplot** summarizes a distribution with **five statistics**: min, Q_1 , Q_2 (median), Q_3 , max, and visually exposes **outliers** and **distributional spread**.

- Q_1 = 25th percentile; Q_3 = 75th percentile.
- $IQR = Q_3 - Q_1$: the interquartile range (box width).
- Whiskers: extend to $Q_1 - 1.5 \cdot IQR$ and $Q_3 + 1.5 \cdot IQR$.

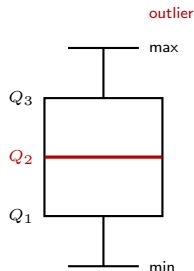


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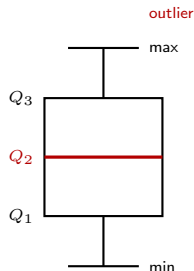


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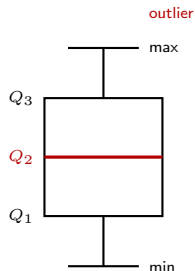


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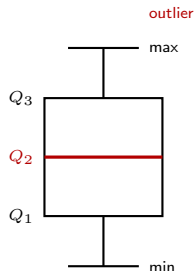


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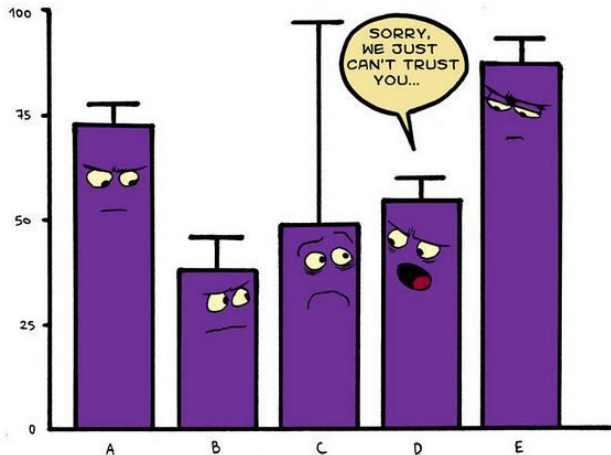
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BoxPlots and Data Confidence II



Correlation Analysis

Correlation

Quantifies the **strength** and **direction** of the **statistical relationship** between two variables. It describes *association*, not **causation**.

- Pearson $r \in [-1, 1]$: linear correlation for continuous, normally distributed variables.
- Spearman ρ : monotonic relationships; rank-based, robust to outliers.
- Kendall τ : based on concordant/discordant pairs; preferred for small samples or many ties.
- Correlation heatmap: color-coded matrix of pairwise coefficients across all variables.

Interpretation Guide

$|r| \geq 0.7$: **strong**. $0.4 \leq |r| < 0.7$: **moderate**. $|r| < 0.4$: **weak** or negligible.

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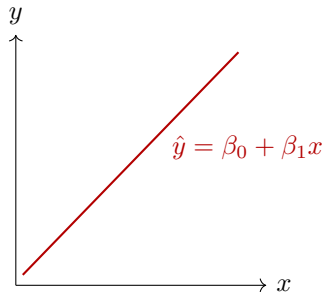
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Tendency Analysis: Linear Regression I

Simple Linear Regression

Linear regression models the **linear relationship** between a dependent variable y and an independent variable x :

$$\hat{y} = \beta_0 + \beta_1 x$$



Tendency Analysis: Linear Regression II

OLS Coefficient Estimates:

$$\beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \beta_0 = \bar{y} - \beta_1 \bar{x}$$

Goodness of Fit:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}, \quad SS_{\text{res}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



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Rust is a **systems programming language** focused on **memory safety**, **zero-cost abstractions**, and **performance**, achieving C-level speed without a garbage collector via its *ownership model*.

Why Rust for data?

- **No runtime overhead**: ideal for **large-scale** data crunching.
- **Fearless concurrency**: safe **parallel** processing with no data races guaranteed at compile time.
- **Polars** core is Rust; Python bindings share the **same speed**.



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Rust for Data Analysis II

Key crates (libraries):

- polars: fast **DataFrame** library backed by **Arrow**.
- arrow2: **Apache Arrow** in-memory columnar format.
- ndarray: **N-dimensional arrays** similar to **NumPy**.
- rayon: **data-parallel** iterators with work-stealing.



Missing Values

Missing Data

Occur when **no data** is recorded for a variable in an observation. They can **bias results**, reduce **statistical power**, and cause **errors** in models if not handled properly.

- **MCAR** — *Missing Completely at Random*: absence is independent of any variable; safest to impute or drop.
- **MAR** — *Missing at Random*: missingness depends on other observed variables (e.g., age missing more often for elderly).
- **MNAR** — *Missing Not at Random*: missingness depends on the missing value itself.



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- **Deletion**: listwise or pairwise *row/column removal*.
- **Mean/Median imputation**: replace with *central tendency*.
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Outlier Detection: IQR and Z-Score

IQR Method

$$IQR = Q_3 - Q_1$$

Bounds:

$$L = Q_1 - 1.5 \cdot IQR$$

$$U = Q_3 + 1.5 \cdot IQR$$

Observations outside $[L, U]$ are **outliers**.

Robust: does **not** assume normality.

Z-Score Method

$$z_i = \frac{x_i - \mu}{\sigma}$$

Values with $|z_i| > 3$ exceed **3 standard deviations** and are flagged as **outliers**.

Assumes: data is **approximately normal**.

When to use which?

Use **IQR** for **skewed** or **non-normal** data. Use **Z-score** when data is **approximately normal** and the **mean** and **standard deviation** are reliable.

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Data Normalization

Feature Scaling

Normalization transforms features to a **common scale**, preventing magnitude-dominated behavior in algorithms sensitive to feature scale such as *KNN*, *SVM*, and *PCA*.

Min-Max Scaling

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Output range: $[0, 1]$.

Sensitive to outliers.

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Output: $\mu' = 0$, $\sigma' = 1$.

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• **Log Transform:** $x' = \log(x + 1)$; reduces right-skewness for heavy-tailed, positive-valued features.



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Pearson Correlation Coefficient I

Pearson's r

Pearson's r measures the **linear** association between two **continuous variables** X and Y , ranging from -1 to $+1$:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \cdot \sum_{i=1}^n (y_i - \bar{y})^2}}$$

- $r = +1$: perfect positive linear relationship.
- $r = -1$: perfect negative linear relationship.
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Polars: Modern DataFrame Library I

Polars

Polars is a **blazingly fast** DataFrame library written in **Rust** with a Python API, built on **Apache Arrow** for columnar **in-memory layout** and supporting **lazy** and **parallel** execution out of the box.

Advantages over pandas:

- **Lazy evaluation**: builds a **query plan** before execution, enabling automatic optimization.
- **Multithreaded**: uses all CPU cores by default with no extra configuration.
- **No index**: simpler mental model, fewer surprises.

Polars also supports file formats like Parquet and CSV, and has a powerful SQL interface for querying data.



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Polars: Modern DataFrame Library II

Key operations:

- `pl.scan_parquet()`: **lazy** Parquet scan; reads only required **columns and rows**.
- `.filter()` / `.select()`: efficient **predicate pushdown** to the storage layer.
- `.group_by().agg()`: fast **group aggregation** exploiting **SIMD** instructions.
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Outline

1 Descriptive Analytics

2 Data Analysis



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/data-analysis-programming>

